SonicSurface 1.2 · 8×8 transducer module

Array8x8 RevC: 64-channel ultrasonic transducer driver board

This report describes the RevC board and the fabrication files generated from it: how it works, its electrical limits, how it is built, what has been verified, and how to bring it up. It is written for engineers who did not design the board. Figures marked *(estimate)* are first-order calculations from datasheet values, not measurements.

DOCUMENT ITEM	VALUE
Board	Array8x8_RevC
Board file	Array8x8_RevC.kicad_pcb, saved 2026-11-04 12:23
Fabrication files	production_JLC/ (Gerber/drill zip, BOM, placement list, IPC netlist), generated 2026-11-04 12:27
Schematic revision	C – UNVERIFIED (title block)
Report date	2026-11-04

AT A GLANCE	VALUE
Function	64-element phased array: 8 × 8 Murata MA40S4S, 40 kHz
Drive supply	16.8 V (VDRV), external, via J2 Molex Micro-Fit 3.0
Logic supply	3.3 V from the controller, via J1 pin 1
Control	8 data lanes + 4 control lines on a 26-pin keyed IDC ribbon, ≤ 100 mm
Board	126.75 × 132.75 mm, 2-layer FR-4, 1.6 mm, 1 oz copper
Parts	289 parts: 182 SMD fitted by JLCPCB, 107 through-hole fitted by hand

Status before ordering

- **Open, fabrication:** via annular rings are 0.15 mm (755 vias) and 0.175 mm (372 vias). JLCPCB's stated absolute minimum is 0.18 mm, and its recommendation is 0.25 mm (section 7).
- **Open, cosmetic:** 45 track corners are sharper than 90°. The sharpest is 26.6°.
- **Not yet done:** no board of this revision has been built or powered. The schematic title block still reads "C – UNVERIFIED", and it refers to a QA_CHECKLIST.md that is not in the project folder.

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1 Block diagram and how it works

An FPGA on a separate controller board streams bit patterns over a ribbon cable. Shift registers turn them into 64 logic-level outputs, MOSFET drivers raise each one to 16.8 V, and each output drives one transducer through a 22 Ω resistor.

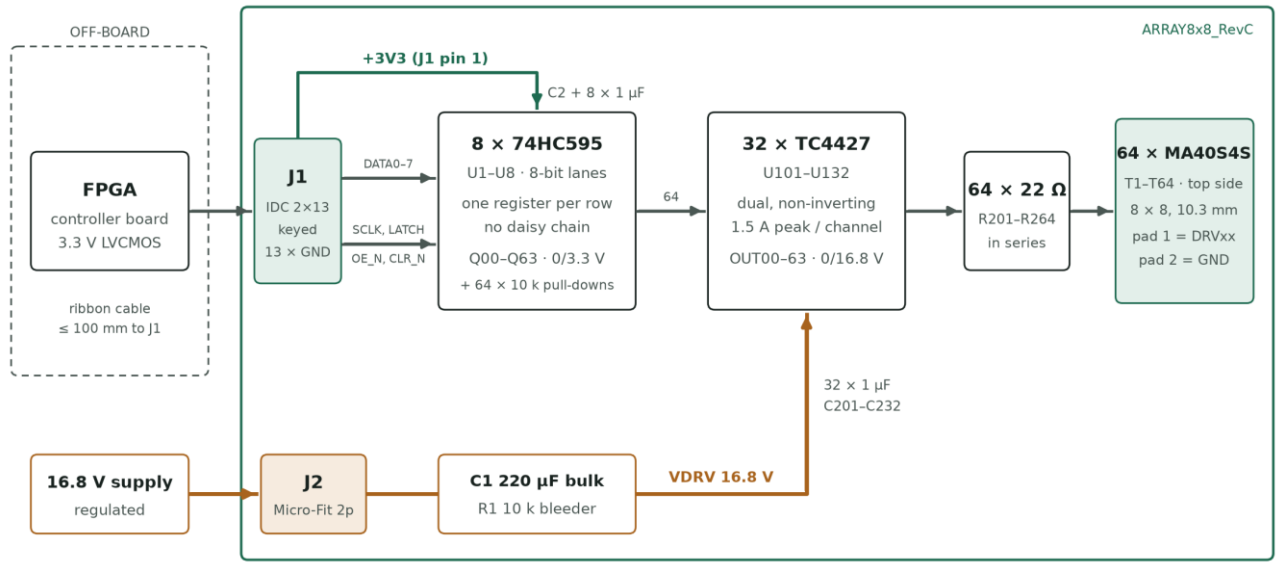


Figure 1. Signal and power paths. Every row of eight transducers has its own register, data line and four drivers, so the chain repeats eight times. The three control lines go to all eight registers.

How signals travel

The 64 channels form 8 identical lanes, one per transducer row. For row k (0–7) the chain is:

```
DATAk_RX (J1) → U(k+1) 74HC595 SER → Q(8k)…Q(8k+7) → 4 × TC4427 → OUT → 22 Ω → DRV → T(8k+1)…T(8k+8)
Row 1 example: DATA0_RX → U1 → Q00–Q07 → U101–U104 → R201–R208 → T1–T8
```

- **No daisy chain.** Each register's serial output (QH', pin 9) is unconnected. The eight registers load in parallel from eight data lines, so a full 64-bit frame takes only 8 clock pulses.
- **Bit order.** The first bit shifted into a lane ends up on QH (Q(8k+7)) after eight clocks. Send the bit for T(8k+8) first and the bit for T(8k+1) last.
- **Frame update.** After 8 SCLK rising edges, one LATCH rising edge copies all 64 bits to the outputs at the same instant. The drivers follow within about 40 ns.
- **Phased array.** Each transducer receives a 40 kHz square wave. The FPGA sets each element's phase by choosing when that element's bit changes. If a 40 kHz period is divided into N time steps, the board needs N × 40 kHz latches per second and 8 SCLK pulses per latch. The firmware chooses N; the board does not fix it.

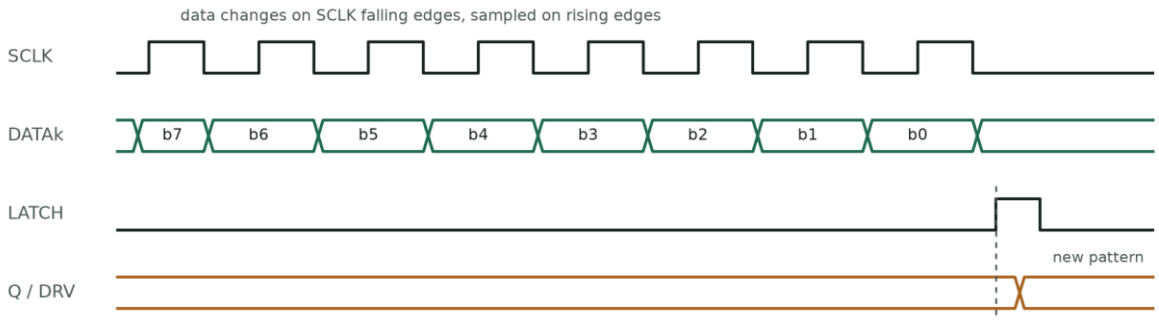


Figure 2. One frame on one lane. All eight lanes shift in parallel on the shared SCLK. b7 lands on QH (column 8) and b0 on QA (column 1). The SCLK↑→LATCH↑ setup time is in section 2.

Power

RAIL	SOURCE	FEEDS	LOCAL PARTS	TP
VDRV 16.8 V	External supply, J2 pin 1	VDD (pin 6) of all 32 TC4427 drivers	C1 220 µF bulk near J2; R1 10 k bleeder; one 1 µF per driver (C201–C232)	TP3
+3V3	Controller board, J1 pin 1	VCC (pin 16) of U1–U8; OE_N pull-up R31	C2 1 µF at J1; one 1 µF per register (C10–C17), 4.4 mm from VCC	TP2
GND	J2 pin 2 and J1 even pins 2–26 (13 pins)	Everything; transducer pad 2	Copper pour on both layers, 1,059 stitching vias	TP1

Drive ground and logic ground are one net on this board. Current returning from the drivers flows mainly to J2, but part of it can also return through the 13 ground wires in the ribbon cable to the controller. Section 9 covers what this means for cabling.

Control signals

SIGNAL	J1 PIN	74HC595 PIN	WHAT IT DOES	DEFAULT IF UNDRIVEN
`DATA0–7_RX`	3, 5 … 17	14 SER (one per register)	Serial data for one row. Sampled on each SCLK rising edge.	Low, 10 k pull-down (R20–R27)
`SCLK_RX`	19	11 SRCLK (all)	Shift clock. Each rising edge shifts every lane by one bit.	Low, R28 pull-down
`LATCH_RX`	21	12 RCLK (all)	Rising edge copies the shift registers to the output registers, updating all 64 channels at once.	Low, R29 pull-down
`OE_N_RX`	23	13 OE (all)	Active-low output enable. High puts all Q outputs in high impedance, so the 10 k pull-downs hold every driver input low and every transducer at 0 V.	**High** (outputs off), R31 pull-up to +3V3
`CLR_N_RX`	25	10 SRCLR (all)	Active-low clear of the shift registers only. The output registers keep their value until the next LATCH.	**Low** (clear held), R30 pull-down

> **Safe default state**
>
> With the cable unplugged or the FPGA unconfigured, OE_N is pulled high and every other line is pulled low. All transducers stay at 0 V. To run, firmware must drive CLR_N high before shifting data, and drive OE_N low only once valid data has been latched.

2 Electrical specifications

Supply limits come from the schematic and the part datasheets. Current and power figures model each transducer as its 2,550 pF datasheet capacitance. Near resonance the real element also draws current through its mechanical branch, so measure supply current at bring-up before relying on these numbers.

Supply voltages and limits

ITEM	DESIGN VALUE	PART LIMIT	MARGIN
VDRV vs. TC4427 supply	16.8 V	4.5–18 V operating; 22 V absolute max	1.2 V to operating max (tight)
Drive amplitude vs. MA40S4S	16.8 Vp-p square	20 Vp-p max (40 kHz square, continuous)	3.2 V (84 % of max)
+3V3 vs. 74HC595	3.3 V	2–6 V recommended	Comfortable
TC4427 input voltage	0–3.3 V	GND − 5 V to VDD + 0.3 V	Exceeded only if logic drives high while VDRV is off; see the power-up order in section 8

16.8 V is also the full-charge voltage of a 4-cell lithium pack. The J2 label "REGULATED DRIVER POWER ONLY" means the input must come from a regulated supply. A supply that overshoots by more than 1.2 V at switch-on runs the drivers above their rated operating range.

Current draw and power

```
Charge per transducer edge   Q = C·V   = 2.55 nF × 16.8 V = 42.8 nC
Average current per element  I = Q·f   = 42.8 nC × 40 kHz = 1.71 mA
Power per element            P = C·V²·f = 28.8 mW   (×64 = 1.84 W, dissipated in drivers and 22 Ω)
```

OPERATING STATE (VDRV)	CURRENT	POWER	BASIS
Idle: OE_N high, all outputs low	≈ 15 mA	0.24 W	32 × 0.4 mA driver supply current (inputs low, datasheet max) + 1.7 mA bleeder
Running: 64 channels at 40 kHz, 50 % duty	≈ 190 mA *(estimate)*	3.2 W	110 mA capacitive + 78 mA driver supply (average of 0.4 and 4.5 mA max) + bleeder
Upper bound	≈ 255 mA *(estimate)*	4.3 W	Capacitive load + 32 × 4.5 mA (all inputs high, datasheet max)

Peak current while switching

QUANTITY	VALUE	HOW IT WAS FOUND
Peak current, one channel	0.58 A	16.8 V ÷ (22 Ω + 7 Ω TC4427 output, typical); 0.52 A with the 10 Ω maximum. The driver is rated 1.5 A peak.
Without the 22 Ω resistor	2.4 A	16.8 V ÷ 7 Ω. This would exceed the 1.5 A rating.
Edge time constant	74 ns	29 Ω × 2.55 nF. The 10–90 % edge is about 160 ns.
All 64 switching on the same edge	≈ 37 A for ~0.2 µs *(estimate)*	64 × 0.58 A, a total of 2.74 µC per rising edge. This is the worst case (every element in phase). Falling edges discharge to ground and draw nothing from VDRV.
Driver package dissipation	≤ 0.09 W	2 × 7 mW switching (its 7/29 share) + up to 76 mW supply current, against a 730 mW PDIP rating
22 Ω resistor dissipation	22 mW	22/29 share of 28.8 mW, against a 100 mW 0603 rating

+3V3 load

The registers draw at most 80 µA each when static. Each Q output that is high also feeds its 10 k pull-down, 0.33 mA each and up to 21 mA with all 64 high. Clocking adds about 5 mA per register at 4 MHz, based on the datasheet's 400 pF power-dissipation capacitance. Plan on **≤ 70 mA** from the controller's 3.3 V output *(estimate)*.

Logic levels

INTERFACE	DRIVER OUTPUT	RECEIVER THRESHOLD	NOTES
FPGA → 74HC595 (VCC 3.3 V)	3.3 V LVCMOS	V_{IH} ≈ 2.31 V, V_{IL} ≈ 0.99 V	TI specifies 0.7 / 0.3 × VCC at 2 V and 4.5 V but not at 3.3 V; these values are interpolated. The FPGA must deliver ≥ 2.31 V high while driving the 10 k pull-downs (0.33 mA each).
74HC595 → TC4427	≈ 3.3 V / 0 V (lightly loaded)	V_{IH} 2.4 V min, V_{IL} 0.8 V max	About 0.9 V of high-level margin. TC4427 inputs leak only ±1 µA.

Timing (SN74HC595, –40 to 85 °C)

TI publishes timing at 2 V, 4.5 V and 6 V only. At 3.3 V, the 2 V column is the guaranteed bound. Faster operation is probably fine at 3.3 V but needs bench confirmation.

PARAMETER	AT 2 V (BOUND)	AT 4.5 V	DESIGN LIMIT AT 3.3 V
SCLK frequency, max	5 MHz	25 MHz	**≤ 4 MHz**, with data changed on the SCLK falling edge
Clock pulse width (SRCLK, RCLK) high or low	100 ns	20 ns	≥ 100 ns
Setup: DATA before SCLK↑	125 ns	25 ns	≥ 125 ns (half of a 4 MHz period)
Setup: SCLK↑ before LATCH↑	94 ns	19 ns	≥ 94 ns after the 8th SCLK edge
Setup: CLR_N high before SCLK↑	60 ns	12 ns	≥ 60 ns
Hold: DATA after SCLK↑	0 ns	0 ns	0 ns
Input rise/fall time, max	1000 ns	500 ns	≤ 500 ns; easily met over 100 mm of ribbon

At 4 MHz, one frame takes 2 µs plus the latch, so the frame rate tops out near 450 kHz. That allows about 11 phase steps per 40 kHz cycle. Finer phase control needs a faster SCLK, verified on the bench. Driver delay is 40–60 ns max (TC4427, over temperature). That is under 0.9° of a 40 kHz period, so channel-to-channel phase error from the drivers is negligible.

Decoupling strategy

PART	QTY	JOB	SIZING CHECK
1 µF 50 V X7R per TC4427 (C201–C232)	32	Supplies each driver's fast edge current locally, so the 0.58 A edge pulses do not travel back along the VDRV tracks.	Both channels switching draw 85.7 nC, a droop of about 86 mV (0.5 %). X7R loses part of its capacitance at 16.8 V DC bias, so expect roughly 0.1–0.2 V in practice *(estimate)*.
220 µF 35 V (C1)	1	Bulk reservoir at the input. Recharges the 32 ceramics between edges (12.5 µs apart) and covers the inductance of the supply cable.	Rated for 850 mA ripple at 100 kHz with 86 mΩ ESR, which covers the ~0.2 A average draw. With all 64 in phase, the ceramics supply the 2.74 µC edge charge first (≈ 86 mV droop on 32 µF).
1 µF per 74HC595 (C10–C17) + C2 at J1	9	Local decoupling for register switching, plus a reservoir where 3.3 V enters the board.	Placed 4.4 mm from each VCC pin.

What the resistors do

- **22 Ω series, R201–R264 (one per transducer).** Limits peak current to about 0.58 A, inside the driver's 1.5 A rating. Damps ringing between track inductance and the transducer's capacitance. Slows edges to about 160 ns, which reduces EMI. Takes about three quarters of the switching loss off the driver.
- **10 k pull-downs on Q00–Q63, R101–R164.** Hold every driver input low while the register outputs are disabled (OE_N high). Without them, the driver inputs would float and the transducers could switch at random.
- **10 k on the interface lines, R20–R31.** Give every J1 input a defined state with the cable unplugged. DATA, SCLK, LATCH and CLR_N are pulled down; OE_N is pulled up so outputs start disabled.
- **10 k bleeder, R1.** Drains C1 and the 32 ceramics after power-off. The time constant is about 2.5 s, reaching below 1 V in about 7 s. It draws 1.7 mA (28 mW) during operation.

3 PCB stack-up and fabrication specs

ITEM	VALUE	SOURCE
Layers	2 copper (F.Cu, B.Cu)	Board file
Thickness	1.6 mm: 1.51 mm FR-4 core (ε_r 4.5), 2 × 35 µm copper, 2 × 10 µm mask	Board stack-up
Copper weight	1 oz (35 µm), both layers	Board stack-up
Solder mask / silkscreen colour	Not set in the design files; chosen on the order form. Green keeps JLCPCB's 0.10 mm mask-bridge minimum (black and white need 0.13 mm).	JLCPCB capabilities
Surface finish	Not set ("None" in the stack-up); chosen on the order form. HASL lead-free or ENIG both suit 0603/0805 assembly. ENIG gives flatter pads.	Board stack-up
Outline	126.75 × 132.75 mm rectangle (expanded 6.35 mm on every side on 2 Nov)	Edge.Cuts

ITEM	VALUE	SOURCE
Mounting holes	5 × 3.2 mm non-plated (M3). Measured from the top-left board corner: H1 (7.64, 7.64), H2 (119.10, 7.64), H3 (7.64, 125.10), H4 (119.10, 125.10), H5 (61.88, 125.10) mm	Board file
Track widths	0.20 mm signals (minimum), 0.25 mm some control runs, 0.30 mm +3V3 and GND, 0.40–0.80 mm VDRV	Measured
Vias	1,127 total: 755 × 0.60/0.30 mm and 372 × 0.65/0.30 mm (pad/drill); 1,059 are GND	Measured
Plated drills	0.30 mm × 1,127 (vias), 0.80 mm × 386 (DIP pins, C1), 1.00 mm × 26 (J1), 1.02 mm × 2 (J2), 1.20 mm × 128 (transducers)	PTH.drl
Non-plated drills	1.27 mm × 2 (J2 locating pegs), 3.20 mm × 5 (mounting)	NPTH.drl

Design rules compared with JLCPCB

The project's rule file starts from JLCPCB's published minimums. Rules added on 2 and 4 Nov raise them: a 0.18 mm floor on copper clearance and track width, 0.30 mm from pour to other-net pads and vias, and 0.50 mm from through-hole pads to pour and SMD pads. JLCPCB limits below are for 1–2 layer boards with 1 oz copper.

RULE	PROJECT RULE	JLCPCB LIMIT	MEASURED	RESULT
Track width	0.18 mm	0.10 mm	0.20 mm	pass
Track-to-track spacing	0.18 mm	0.10 mm	0.182 mm	pass
Via-to-track	0.18 mm	0.10 mm	0.201 mm	pass
Via-to-via, different nets	0.18 mm	0.20 mm hole-to-hole	0.223 mm (copper)	pass
SMD pad to other-net pad	0.18 mm	0.10 mm	0.495 mm	pass
Via hole / diameter	0.30 / 0.60 mm	0.15 / 0.25 mm min	0.30 / 0.60–0.65 mm	pass
Via annular ring	0.075 mm	0.18 mm abs. min; 0.25 mm recommended	0.15 / 0.175 mm	below JLC
PTH annular ring	0.075 mm	≥ 0.20 mm	0.20 mm (transducer 1.6/1.2 mm pads)	at limit
Hole-to-hole	0.50 / 0.254 mm (diff. / same net)	0.20 mm via, 0.45 mm pad	DRC clean	pass
Copper to board edge	0.30 mm	≥ 0.20 mm (routed)	DRC clean	pass
Solder-mask web	0.10 mm	0.10 mm (green)	No SMD web under 0.18 mm	pass
Silkscreen line width / text height	0.15 / 1.0 mm	0.15 / 1.0 mm	No line under 0.15 mm	pass
Silkscreen to pad	0.15 mm	0.15 mm	1 item (J2's own reference)	minor

The project rule for annular ring (0.075 mm) is looser than JLCPCB's guidance, so DRC does not flag the vias. The fix in the 2 Nov review keeps every via's outer diameter and reduces the drill: 0.60/0.20 mm and 0.65/0.25 mm give 0.20 mm rings with no other copper moving.

4 Layout and routing

Placement plan

- **Top side:** only the 64 transducers, on an 8 × 8 grid at 10.3 mm pitch (centres span 72.1 mm in each direction). Half are rotated 180°. Pad 1 is still the drive pin and pad 2 still ground on every element; only the physical orientation alternates.
- **Bottom side:** everything else. The eight shift registers run in a column at the right edge (x = 140.5 mm), one per transducer row at the same 10.3 mm pitch. Four drivers sit inside each row band, under the transducers they drive. J1, J2, C1, the 22 Ω and 10 k resistors, the decoupling capacitors and all test pads are also on the bottom.
- This keeps the top face flat for the acoustic aperture and puts every machine-placed part on one side, so it is a single-sided SMT job.

Ground pour and return currents

- GND pours fill both layers: 12,144 mm² on F.Cu and 11,738 mm² on B.Cu (filled area), with 0.20 mm minimum width and thermal reliefs (0.25 mm gap, 0.30 mm spokes) on through-hole pads.
- 1,059 GND vias stitch the two pours together across the array. Transducer return current goes from pad 2 straight into the pour, and each driver's switching loop closes through its local 1 µF and the pour.
- Pours are held back 0.30 mm from other-net pads and vias, and 0.50 mm from through-hole pads, to pass JLCDFM's through-hole-to-SMD check.

VDRV routing

VDRV has no pour of its own; it is routed as tracks on B.Cu only.

WIDTH	TOTAL LENGTH	USE	APPROX. CAPACITY, 10 °C RISE (IPC-2221)
0.80 mm	105 mm	Trunk from J2 / C1	≈ 2.0 A
0.60 mm	26 mm	Junctions	≈ 1.6 A
0.50 mm	638 mm	Row distribution runs	≈ 1.4 A
0.40 mm	224 mm	Branches to each driver's pin 6	≈ 1.2 A

Average VDRV current is at most about 0.26 A, well inside these widths. The fast edge current comes from the local 1 µF capacitors. They sit 8.8–10.7 mm from their driver's VDD pin, which adds loop inductance. That is acceptable at 160 ns edges, but in a future revision the capacitors should move next to pin 6.

Layout decisions that matter

CHANGE	WHY	RESULT NOW
R113, R121, R129, R137, R153 and R161 moved away from transducer pads T5–T53	Pads were 0.033 mm apart, a near-short with mask bridges	0.50–0.73 mm for those six; ≥ 0.495 mm for every SMD pad on the board
C224, C228 and C232 pad 1 joined to the VDRV branches	Three drivers had no decoupling	All 32 driver capacitors connected; 0 unconnected items
Vias moved out of SMD pads	Via-in-pad wicks solder away from 0603/0805 joints	0 vias inside SMD pads
Pour pulled back from pads (0.30 mm) and through-hole pads (0.50 mm)	JLCDFM through-hole-to-SMD check	Rules in place, DRC clean
17 test pads added on the back (TP10–TP26)	Probe access to each register and each FPGA input	No vias in any test pad
Outline grown 6.35 mm per side; fifth mounting hole H5	Mechanical margin and support along the connector edge	126.75 × 132.75 mm, 5 × M3

5 Bill of materials

Quantities are per board. The project builds two boards, so double every quantity. JLCPCB part pages were checked on 4 Nov 2026; those pages do not show live stock.

REFERENCES	QTY	VALUE	FOOTPRINT	LCSC / MANUFACTURER PART	FITTED BY
C2, C10–C17, C201–C232	41	1 µF 50 V X7R ±10 %	0805	C28323 · Samsung CL21B105KBFNNNE	**JLCPCB · Basic**
R1, R20–R31, R101–R164	77	10 kΩ 1 % 0.1 W	0603	C25804 · UNI-ROYAL 0603WAF1002T5E	**JLCPCB · Basic**
R201–R264	64	22 Ω 1 % 0.1 W	0603	C23345 · UNI-ROYAL 0603WAF220JT5E	**JLCPCB · Basic**
C1	1	220 µF 35 V, 86 mΩ, 850 mA ripple	Radial D8 P3.5	C106697 · AISHI ERS1VM221F12OT	**Hand**
U1–U8	8	8-bit shift register, 3-state outputs	DIP-16	TI SN74HC595N	**Hand**
U101–U132	32	Dual non-inverting MOSFET driver, 1.5 A	DIP-8	Microchip TC4427EPA	**Hand**
T1–T64	64	40 kHz ultrasonic transmitter, 10 mm	2-pin, 5 mm	Murata MA40S4S	**Hand**
J1	1	2 × 13 keyed IDC header, 2.54 mm, vertical	IDC-Header_2x13	Any keyed (shrouded) 26-pin box header	**Hand**
J2	1	2-pin power header, 3.00 mm, vertical, pegs	Micro-Fit 3.0	Molex 43650-0215	**Hand**
TP1–TP6, TP8–TP26	25	1.5 mm test pads	Bare copper	None (pads only)	—
H1–H5	5	M3 mounting holes	3.2 mm NPTH	None	—

JLCPCB's order files (BOM and placement list) cover only the 182 bottom-side SMD parts. C1 is in the JLCPCB library as an Extended through-hole part, but it is fitted by hand with the other through-hole parts.

Alternates and stock risks

PART	RISK	ALTERNATES
MA40S4S	**Check.** 128 needed for two boards. Murata's product page returned "not found" on 4 Nov 2026. Confirm lifecycle status and distributor stock before ordering.	No drop-in alternate checked. Any substitute must match 40 kHz, the 10 mm body, the 5 mm pin pitch and ≥ 20 Vp-p drive. Phase matching between elements matters more than brand.
TC4427EPA	Low. Mature PDIP part; 64 for two boards.	TC4427AEPA (Microchip's same-pinout "A" revision). Check its datasheet for any change in VDD range or input thresholds.
SN74HC595N	Low. Multi-source logic.	74HC595N from Nexperia, onsemi or others. 74AHC595 in DIP has more timing margin at 3.3 V where it is available.
0603 / 0805 passives	Low. JLCPCB Basic parts, kept loaded.	Any 1 % thick-film 0603; any 1 µF ≥ 25 V X7R 0805 (50 V keeps more capacitance at 16.8 V bias).
Molex 43650-0215	Low. Distributor stock in the tens of thousands on 4 Nov 2026.	43650-0216 / -0217, per the footprint's own notes.

6 Assembly notes

Who fits what

- **JLCPCB (single-sided SMT, bottom):** 141 resistors and 41 capacitors. Before ordering, check JLCPCB's placement preview: every part should sit on its pads.
- **By hand (through-hole):** 40 DIP ICs, 64 transducers, C1, J1 and J2.

Hand-assembly order

1. Inspect the SMT parts from JLCPCB. Check for bridges on the 22 Ω resistors next to the transducer pads.
2. Fit the 40 DIP ICs on the **bottom**, with pin 1 matching the silkscreen, and solder them from the top.
3. **Trim every DIP lead flush on the top side.** DRC finds 199 driver and register lead tips inside transducer outlines, so untrimmed leads would hold the transducers off the board or touch their cases. Do not use IC sockets unless their pin length has been checked against this.
4. Fit C1 (12 mm tall), J2 and J1 on the bottom. Check J1's key slot orientation against the cable before soldering.
5. Fit the 64 transducers last, from the top, and solder from the bottom. Keep iron contact short.
6. Clean flux, then do the unpowered checks in section 8.

Transducer polarity and orientation

- Footprint pad 1 is always the drive pin (DRVxx) and pad 2 is always ground, but half the footprints are rotated 180°. Follow each position's silkscreen marking; do not insert all 64 the same way round.
- One MA40S4S lead connects to the metal case. Confirm which with a continuity meter before fitting, and put that lead on the ground pad (pad 2) for all 64.
- A reversed element runs 180° out of phase with its neighbours and partly cancels the array's output, so get polarity right before soldering. Swapping a soldered transducer is slow.

7 Design verification

All checks below ran on 4 Nov 2026 against the board file saved at 12:23, using KiCad 10 DRC with schematic parity plus direct geometry measurements.

DRC and schematic-to-board check

CHECK	RESULT	MEANING
Unconnected items	**0**	Every net is fully routed
Clearance, short and width errors (incl. 0.18 mm floor)	**0**	Copper meets the project rules
Courtyard overlaps	**87 errors**	85 are 0603 resistors placed tight against DIP courtyards, and 2 are capacitor-to-resistor. SMT goes on before the DIPs, so these are accepted.
Silkscreen overlap	**1 error**	J2's outline crosses its own reference text. Cosmetic.
Through-hole pin inside courtyard	**199 warnings**	DIP leads under transducers; covered by the lead-trim step in section 6
Footprint differs from library	**199 warnings**	Deliberate pad and courtyard edits in the board copy
Schematic parity	**no net mismatch**	85 warnings: 72 unused pins (e.g. QH' pin 9, TC4427 NC pins), 8 test-pad BOM-flag differences, 5 mounting holes not in the schematic
Fabrication files vs. board	**match**	Gerber/drill zip, BOM and placement list regenerated at 12:27 from the 12:23 board. Placement coordinates match the pad centres exactly.

DFM review findings (from 2 Nov, re-measured 4 Nov)

#	ITEM	2 NOV	NOW	STATUS
1	Pad-to-pad near-short (resistors vs. transducers)	0.033 mm, 5 places	≥ 0.495 mm	**fixed**
2	Driver decoupling not connected	3 open	0 open	**fixed**
3	Via annular ring	0.15 / 0.175 mm	0.15 / 0.175 mm	**open**
4	Track-to-track spacing	0.127 mm, 19 places	0.182 mm min	**fixed**
5	Via-to-track	0.155 mm, 30 places	0.201 mm min	**fixed**
6	Track width	0.15 mm, 14 segments	0.20 mm min	**fixed**
7	Via-to-via, different nets	0.175 mm	0.223 mm	**fixed**
8	Via in SMD pad	7 pads	0	**fixed**
9	Acute track corners	6 corners	45 junctions under 90° (smallest 26.6°), on DATA0/4/5/6/7 near J1 and on Q32. The count is higher because this measurement includes every junction under 90°.	**open, low**
10	Silkscreen line width	0.10 mm	none under 0.15 mm	**fixed**
11	Silkscreen to pad	10 places	1 (J2)	**mostly fixed**

#	ITEM	2 NOV	NOW	STATUS
12	Assembly fit (courtyards, leads under transducers)	87 / 199	87 / 199	accepted
13	PTH annular ring	0.20 mm	0.20 mm	at JLC limit
14–16	Holes, mask webs, edges, outline	pass	pass	pass

Known issues and accepted risks

ISSUE	RISK	DISPOSITION
Via rings below JLCPCB's 0.18 mm minimum	Fab: drill breakout, intermittent vias	Recommended before ordering: reduce via drills to 0.20 / 0.25 mm and re-run DRC and the fab export
VDRV 16.8 V vs. TC4427 18 V operating max	Electrical: overshoot at switch-on	Regulated supply only; check switch-on overshoot with a scope
No fuse, reverse-polarity or transient protection on VDRV	Electrical: a miswired supply damages 32 drivers	Accepted. J2 is polarized and keyed; consider an inline fuse in the supply lead.
Driver decoupling 9–11 mm from VDD pins	Signal: more supply ringing at edges	Accepted for RevC; move the capacitors closer in the next revision
Drive and logic share one ground through the ribbon	Signal: ground bounce at the controller	Keep the ribbon ≤ 100 mm; join the supply ground to the controller ground at one point
Current figures are first-order	Planning: real draw near resonance may be higher	Measure at bring-up (section 8) and resize the supply if needed
Schematic marked unverified; QA checklist missing	Process	Write the checklist from section 8 and clear the flag after the first board passes
MA40S4S availability	Supply	Confirm stock and lifecycle before committing to two boards

8 Testing and bring-up

Test-point map

All test pads are 1.5 mm bare copper on the bottom. There is no TP7; the numbering skips it.

TP	NET	IDLE / CABLE UNPLUGGED	RUNNING
TP1	GND	Reference for every measurement	Reference
TP2	+3V3	3.3 V (0 V if the controller is off)	3.3 V ± 5 %
TP3	VDRV	16.8 V	16.8 V; ripple at 40 kHz
TP4	SCLK_RX	0 V (pull-down)	Bursts of 8 clock pulses
TP5	LATCH_RX	0 V	One pulse per frame
TP6	OE_N_RX	3.3 V (pull-up: outputs off)	0 V (outputs on)
TP8	DRV00	0 V	0 / 16.8 V square, 40 kHz (T1)
TP9	DRV63	0 V	0 / 16.8 V square, 40 kHz (T64)
TP10–TP17	Q00, Q08 … Q56	0 V (pull-down)	0 / 3.3 V; column 1 of rows 1–8

TP	NET	IDLE / CABLE UNPLUGGED	RUNNING
TP18–TP25	DATA0–7_RX	0 V	Serial data, changing on SCLK falling edges
TP26	CLR_N_RX	0 V (clear held)	3.3 V

Power-up sequence and first checks

1. **Unpowered.** Check that neither VDRV–GND nor +3V3–GND is a short. VDRV reads a rising resistance as the capacitors charge from the meter, settling near the 10 k bleeder. Confirm J1 pin 1 is +3V3 and the even pins are GND.
2. **VDRV first, current-limited.** Apply 16.8 V from a bench supply limited to about 0.3 A, with the cable unplugged. Expect about 15 mA: 32 drivers at their 0.4 mA idle supply current plus the bleeder. TP8 and TP9 should read 0 V. Much higher current means a reversed DIP or a short.
3. **Then logic.** Connect the controller with OE_N held high, then power +3V3. Check TP2 and the idle states above. Applying VDRV before logic keeps the TC4427 inputs from ever exceeding VDD + 0.3 V.
4. **Power-down in reverse:** OE_N high, then logic off, then VDRV off. Wait about 7 s for the bleeder before handling the board.

Functional tests with known patterns

1. **Clear and enable.** Drive CLR_N high and shift in all zeros. Latch, set OE_N low, and confirm every DRV is 0 V.
2. **Walking one.** Load a single 1 into one lane position, latch, and confirm that only the matching transducer's drive pin reads 16.8 V. Step through all 64 positions. Probe at the transducer lead or the 22 Ω pad, and use TP8, TP9 and TP10–TP17 as reference points. This proves the row and column mapping and the bit order (first bit in → column 8).
3. **All on, static.** Load all ones and measure VDRV current. Expect it to stay at or below 32 × 4.5 mA = 144 mA plus the bleeder.
4. **40 kHz, low voltage first.** Run all channels in phase at 40 kHz with VDRV set to 5–8 V. Check supply current and driver temperature, then raise to 16.8 V while comparing the current against the ≈ 190 mA estimate.
5. **Phase check.** Program a known phase ramp across a row. On a two-channel scope, confirm the time offset between DRV00 (TP8) and other channels matches the programmed steps. Confirm acoustic output with a 40 kHz receiver or microphone.
6. **Safety lines.** While running, drive OE_N high and confirm all outputs drop to 0 V. Pulse CLR_N low followed by a latch, and confirm all outputs go low.

9 Interfaces and connectors

J1: control and logic power (26-pin keyed IDC)

A 2 × 13, 2.54 mm vertical box header on the bottom side. Use a keyed (shrouded) header so the cable cannot be fitted reversed. Every odd-numbered signal pin sits next to a ground pin, so each signal has its own return wire in the ribbon. Keep the cable **≤ 100 mm**, as marked on the board: short ribbon keeps crosstalk, edge ringing and ground offset between the boards small at multi-MHz clock rates.

PIN	SIGNAL	PIN	SIGNAL
1	+3V3 (in)	2	GND
3	DATA0_RX → U1	4	GND
5	DATA1_RX → U2	6	GND
7	DATA2_RX → U3	8	GND
9	DATA3_RX → U4	10	GND
11	DATA4_RX → U5	12	GND
13	DATA5_RX → U6	14	GND
15	DATA6_RX → U7	16	GND
17	DATA7_RX → U8	18	GND
19	SCLK_RX	20	GND
21	LATCH_RX	22	GND
23	OE_N_RX	24	GND
25	CLR_N_RX	26	GND

All signals are inputs to this board at 3.3 V LVCMOS. The board returns nothing to the controller. +3V3 arrives on pin 1 only; budget ≤ 70 mA for it (section 2).

J2: drive power input (Molex Micro-Fit 3.0)

ITEM	VALUE
Connector	Molex 43650-0215, Micro-Fit 3.0 vertical header, 1 × 2, 3.00 mm pitch, PCB polarizing pegs, through-hole
Pinout	Pin 1 = VDRV (+16.8 V), pin 2 = GND
Contact rating	8.5 A max per contact and 600 V (Molex). The real limit depends on the mating terminal and wire gauge.
Mating part	Micro-Fit 3.0 2-circuit receptacle housing with crimp terminals, or a ready-made Micro-Fit cable (Molex lists 2147501021, 150 mm)
Input requirement	16.8 V regulated, staying below 18 V including switch-on overshoot. At least 0.5 A capability, which leaves 2× margin over the ≈ 0.26 A upper estimate. Confirm with the bring-up current measurement.
On-board protection	None beyond the keyed, polarized housing: no fuse, reverse-polarity diode or TVS

Sources

- Board and schematic: Array8x8_RevC.kicad_pcb (saved 2026-11-04 12:23), *.kicad_sch, Array8x8_RevC.kicad_dru, DFM_ROUTING_REVIEW.md (2 Nov 2026).
- Fabrication files: production_JLC/ (Gerber/drill zip, BOM, placement list, IPC netlist), generated with Fabrication Toolkit 5.3.1.
- Texas Instruments SN74HC595 datasheet, SCLS041J (rev. Oct 2021): ti.com/lit/ds/symlink/sn74hc595.pdf
- Microchip TC4426/TC4427/TC4428 datasheet, DS20001422G: ww1.microchip.com/downloads/en/DeviceDoc/20001422G.pdf

- Murata MA40S4S/MA40S4R datasheet (distributor copies): 40 kHz, 2,550 pF ±20 % at 1 kHz, 120 ±3 dB SPL at 30 cm (10 Vrms), max input 20 Vp-p square continuous.
- JLCPCB PCB capabilities (jlcpcb.com/capabilities/pcb-capabilities) and part pages C28323, C25804, C23345, C106697 (jlcpcb.com/partdetail/…).
- Molex 43650-0215 product page: molex.com/en-us/products/part-detail/436500215

Estimates are marked; everything else was measured from the board file or taken from the cited datasheets.